

Task 2 N3AS

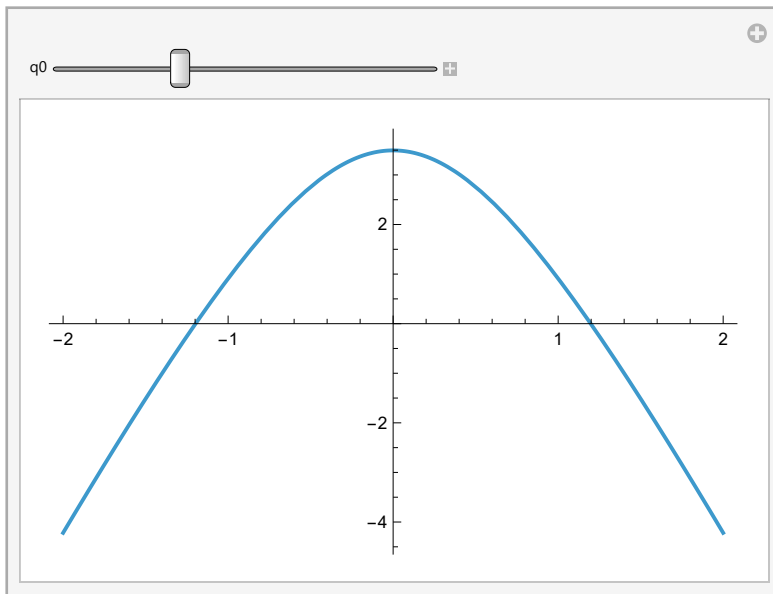
Plots

Plots For Positive Roots

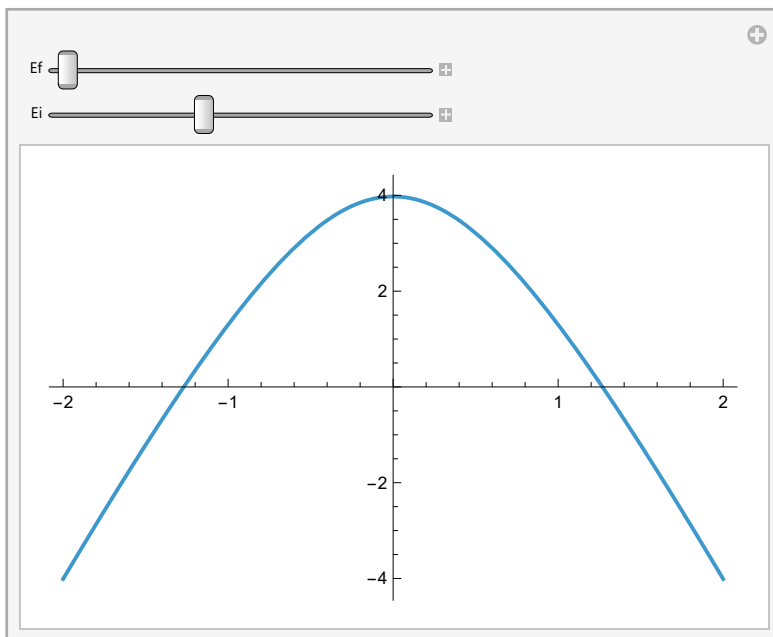
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In[1]:= f1[k_, q0_] := q0^2 - 2 q0 Sqrt[k^2 + 1]
f2[k_, Ei_, Ef_] := (Ei - Ef)^2 - 2 (Ei - Ef) Sqrt[k^2 + 1]
f3[k_, Ef_] := (Ef)^2 + 2 (Ef) Sqrt[k^2 + 1]
f4[k_, q0_] := q0^2 + 2 q0 Sqrt[k^2 + 1]
f5[k_, Ei_, Ef_, q0_] := (Ei - Ef - q0)^2 - 2 (Ei - Ef - q0) Sqrt[k^2 + 1]
f6[k_, Ef_, q0_] := (q0 + Ef)^2 + 2 (q0 + Ef) Sqrt[k^2 + 1]
f7[k_, Ei_, Ef_] := (Ef - Ei)^2 - 2 (Ef - Ei) Sqrt[k^2 + 1]
f8[k_, Ei_, Ef_, q0_] := (q0 + Ef - Ei)^2 - 2 (q0 + Ef - Ei) Sqrt[k^2 + 1]
f9[k_, Ei_] := (Ei)^2 + 2 (Ei) Sqrt[k^2 + 1]
f10[k_, Ef_] := (Ef)^2 - 2 (Ef) Sqrt[k^2 + 1]
f11[k_, Ef_, q0_] := (q0 + Ef)^2 - 2 (q0 + Ef) Sqrt[k^2 + 1]
f12[k_, Ei_] := (Ei)^2 - 2 (Ei) Sqrt[k^2 + 1]

In[13]:= Manipulate[Plot[f1[k, q0], {k, -2, 2}], {q0, 0, 10}]
Manipulate[Plot[f2[k, Ei, Ef], {k, -2, 2}], {Ef, 1.8, 10}, {Ei, Ef, 10}]
Manipulate[Plot[f3[k, Ef], {k, -2, 2}], {Ef, 1.8, 10}]
Manipulate[Plot[f4[k, q0], {k, -2, 2}], {q0, 0, 10}]
Manipulate[Plot[f5[k, Ei, Ef, q0], {k, -2, 2}], {Ef, 1.8, 10}, {Ei, Ef, 10}, {q0, 0, 10}]
Manipulate[Plot[f6[k, Ef, q0], {k, -2, 2}], {Ef, 1.8, 10}, {q0, 0, 10}]
Manipulate[Plot[f7[k, Ei, Ef], {k, -2, 2}], {Ef, 1.8, 10}, {Ei, Ef, 10}]
Manipulate[Plot[f8[k, Ei, Ef, q0], {k, -2, 2}],
  {Ef, 1.8, 10}, {Ei, 1.8, 10}, {q0, 0, 10}]
Manipulate[Plot[f9[k, Ei], {k, -2, 2}], {Ei, 1.8, 10}]
Manipulate[Plot[f10[k, Ef], {k, -2, 2}], {Ef, 1.8, 10}]
Manipulate[Plot[f11[k, Ef, q0], {k, -2, 2}], {Ef, 1.8, 10}, {q0, 0, 10}]
Manipulate[Plot[f12[k, Ei], {k, -2, 2}], {Ei, 1.8, 10}]
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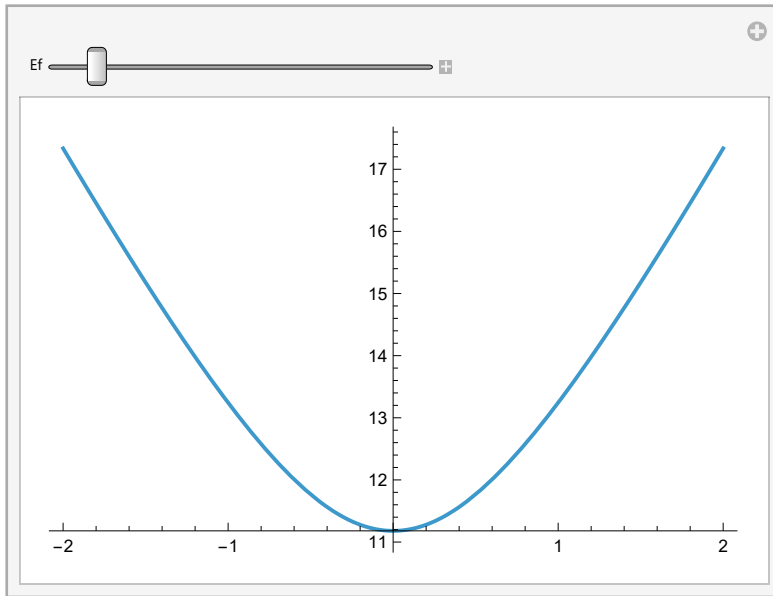
Out[13]=



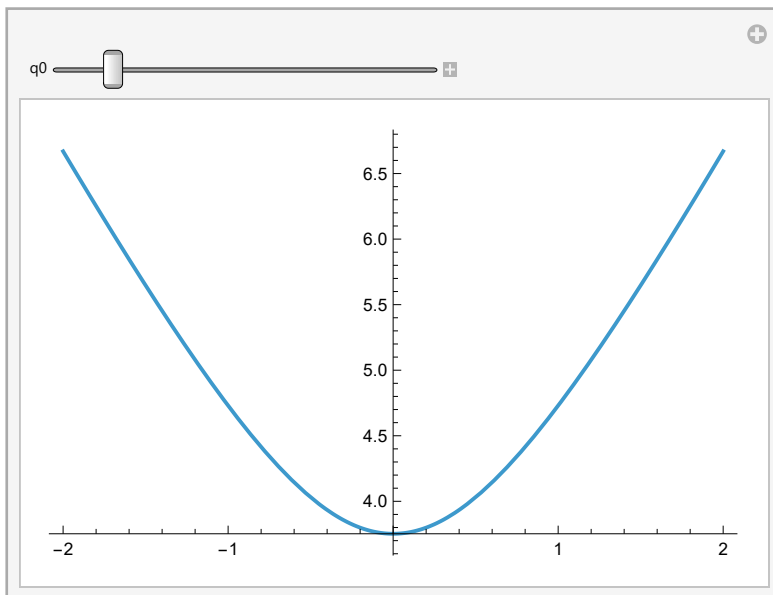
Out[14]=



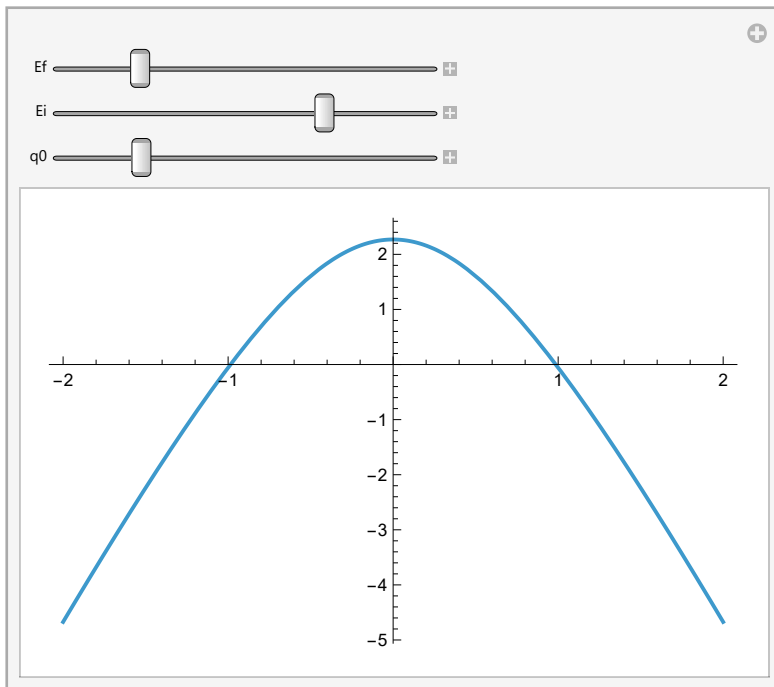
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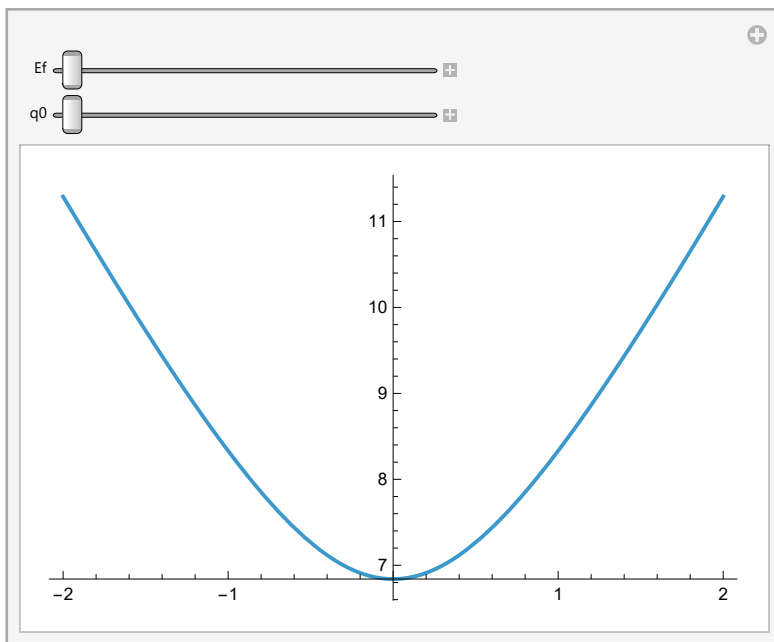
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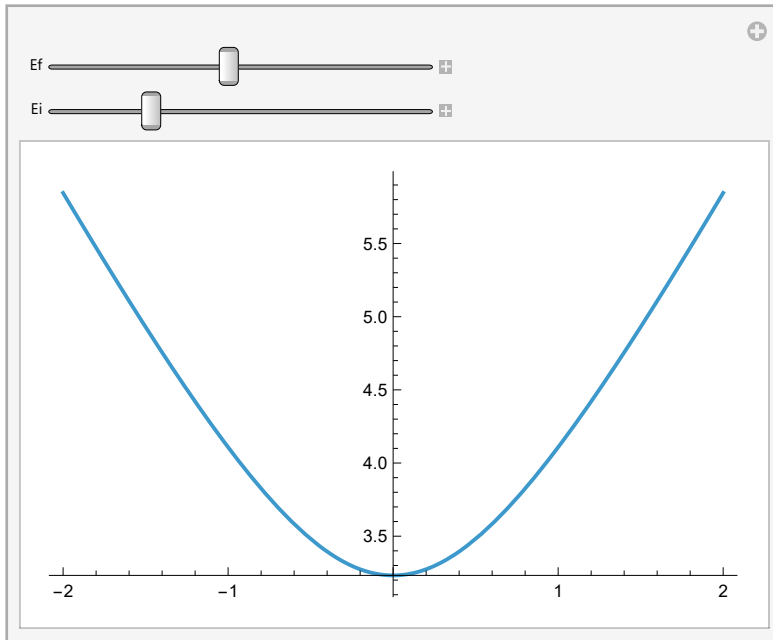
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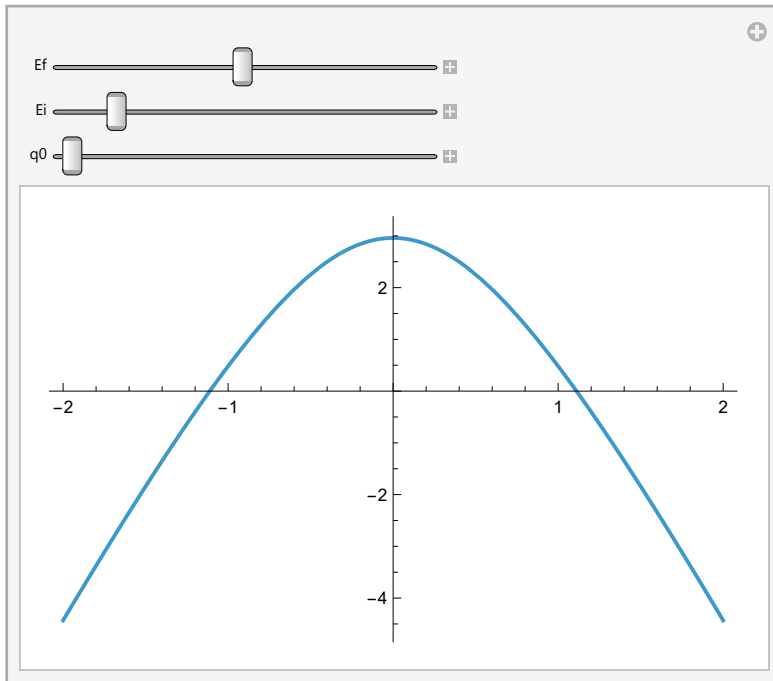
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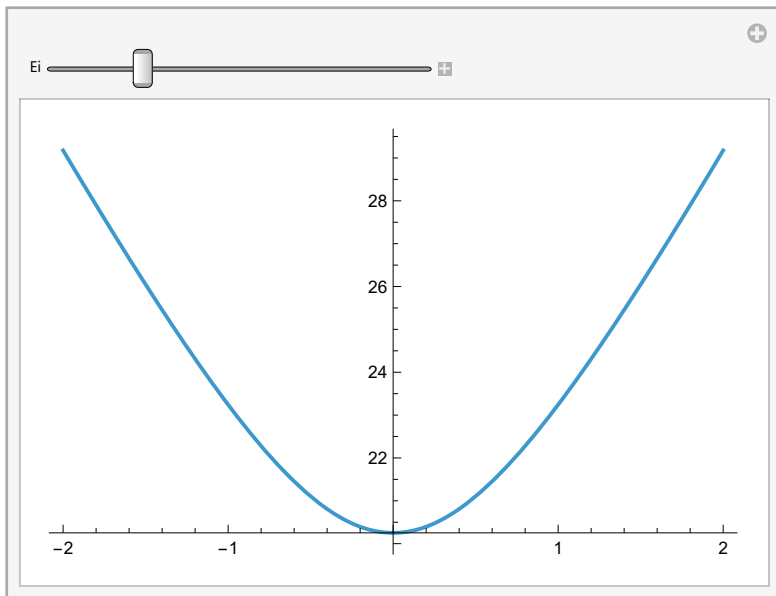
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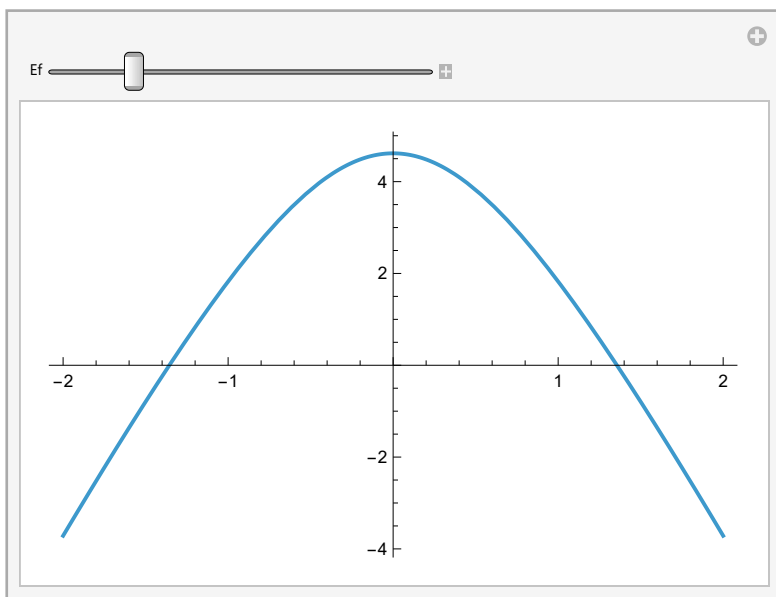
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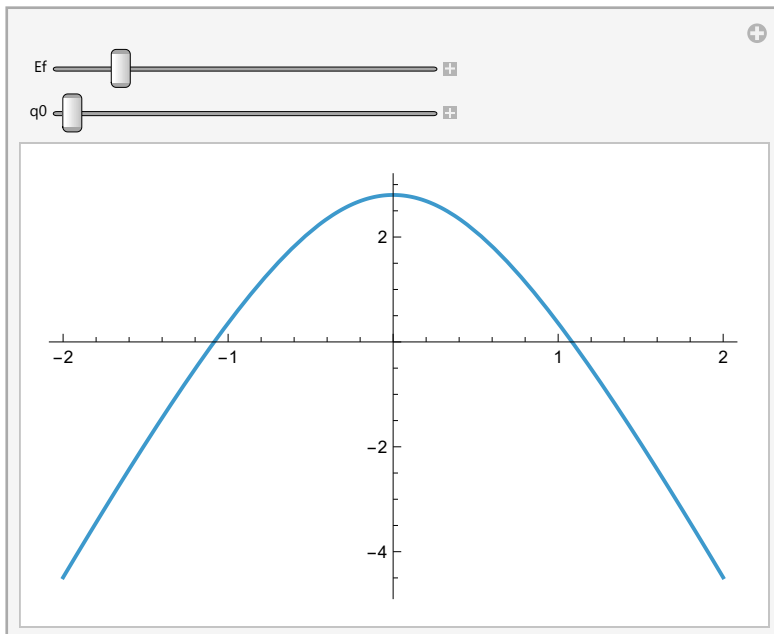
Out[21]=



Out[22]=



Out[23]=



Out[24]=

